MILITARY SPECIFICATION

KIT, MODIFICATION, FOR DELAY TIMER, M1 AND M2

CARGO PARACHUTE RELEASE ASSEMBLIES

This specification is approved for use by the Natick Research and Development Command (GL), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a timer modification kit, which consists of a mounting block assembly, to be used to retrofit time delay assemblies for Releases, Cargo Parachutes, M1 and M2.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard

MILITARY

MIL-P-116 - Preservation - Packaging, Methods of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1670

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-810 - Environmental Test Methods.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.

DRAWING

US ARMY NATICK RESEARCH AND DEVELOPMENT COMMAND

11-1-2660 - Mounting Block Assembly

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval (see 4.3 and 6.3).

3.2 Materials and construction. The materials, construction and supplementary processes involved in the fabrication of the delay timer assembly shall conform to the requirements specified herein and as specified on Drawing 11-1-2660 and all subsidiary drawings referenced thereon.

3.3 Performance.

3.3.1 Timing cycle. The time delay unwinding cycle shall be 12 seconds, plus or minus 3 seconds, from start till completion, and there shall be no evidence of interference in dropping down to its unlocked position after completion of the unwinding cycle when tested as specified in 4.4.2.2.

3.4 Government loaned property. Either one of the items listed below is required for the functional test (see 4.5.1 and figure 1) and may be obtained from the contracting officer on a loan basis (see 6.2).

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Item	Part No.
Release, Cargo Parachute, Airdrop; M-1	11-1-1487
Release, Cargo Parachute, Airdrop; M-2	11-1-565

3.5 Identification of product. Identification marking shall be as specified on the applicable drawings.

3.6 Workmanship. All components and assemblies of the delay timer assembly shall be free from dirt and other extraneous material, burrs, slivers, dents and cracks. External surfaces shall be free from sharp edges and corners. Components shall not be fractured, buckled, loose, dented, missing, malformed or misaligned. Fasteners shall be properly and securely seated.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), the timer modification kit shall be examined for defects in table I and tested as specified in 4.5.1. The presence of any defect or the failure of any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the

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requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.4.2 End item inspection. The lot shall consist of all timer modification kits offered for inspection at one time. The sample unit shall be a complete timer modification kit.

4.4.2.1 Visual examination. Examination shall be made of the timer modification kit for the defects in table I. The inspection level shall be III and the acceptable quality level (AQL) shall be 1.0 for major defects and 4.0 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Omitted or not as specified	X	
	Any evidence of corrosion	X	
	Dirt, oil, grease, or other foreign matter		X
Construction and workmanship	Any component cracked, malformed, misaligned or damaged	X	
	Sharp edges, burrs, or metal slivers		X
	Components incorrectly assembled, loose or missing	X	
	Fasteners improperly or not securely seated		X
Identification marking	Missing, incomplete, illegible, improper location or not as specified		X

4.4.2.2 End item testing. Testing shall be performed as specified in 4.5.1 for conformance to the requirements specified in 3.3.1. The sample unit shall be a complete timer modification kit. The inspection level shall be S-1 with an AQL of 4.0 expressed in terms of percent defective.

4.4.3 Packaging inspection. An examination shall be made to determine that the preservation, packing and marking comply with section 5 requirements. The sample unit shall be one shipping container fully packaged, except that it need not be closed. Characteristics to be examined and the defects applicable thereto shall be as listed in table II. Examination for closure defects shall be made on shipping containers fully packaged. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

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TABLE II. Packaging inspection

Examine	Defect
Marking, interior and exterior containers	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Any component missing or not as specified. Any component damaged affecting serviceability.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, inadequate stapling, or improper taping. Bulged or distorted container.
Content	Number of kits per shipping container is more or less than required.

4.5 Methods of inspection.

4.5.1 Delay timer unwinding cycle. The delay timer assembly of the cargo parachute release provided as GLP (see 3.4) shall be modified by replacing its mounting block, and timer and winding stem with the Timer Modification Kit. Testing of the delay timer assembly shall be conducted in a temperature environment conforming to Procedure I of Methods 501 and 502 of MIL-STD-810. The modified delay timer assembly shall be tested with the stem and latches in a horizontal position and held in a cargo parachute release; type M-1 or M-2 (GLP, see 3.4), in the manner shown on figure 1. An additional weight shall be suspended from the modified delay timer assembly using some suitable textile or metal sling (see figure 1). The stem shall be fully wound in a clockwise direction approximately 90 degrees, engaging the latches in the release cavity slots (locked position), and the arming wire inserted. The modified assembly shall then be actuated by removal of the arming wire and the length of the unwinding cycle shall be measured by a suitable time lapse instrument. After the unwinding cycle has been completed, the modified delay timer assembly shall drop freely and smoothly within the release to its lower unlocked position. Failure of the modified assembly to meet either the time requirement of 3.3.1 or any indication of interference in dropping into the unlocked position, shall constitute a defect.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 2.1).

5.1.1 Level A.

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5.1.1.1 Unit pack. Each timer modification kit shall be provided mechanical and physical protection in accordance with method III of MIL-P-116. When cushioning is used as a protective measure it shall be of the non-dusting type.

5.1.1.2 Intermediate pack. Ten timer modification kits, unit packed as specified, shall be placed in a snug-fitting corrugated fiberboard box. The fiberboard box shall conform to style RSC or CSSC, type CF, variety SW, class domestic, minimum grade 125 of PPP-B-636. The fiberboard box shall be closed with tape as specified in PPP-B-636.

5.1.2 Commercial. Timer modification kits shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B or Commercial, as specified (see 6.2).

5.2.1 Level A packing. One hundred and twenty timer modification kits, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each fiberboard container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of PPP-B-636.

5.2.2 Level B packing. One hundred and twenty timer modification kits, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V3c, V3s or V4s of PPP-B-636. Each fiberboard container shall be closed in accordance with method III as specified in the appendix of PPP-B-636.

5.2.3 Commercial. One hundred and twenty timer modification kits, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking. In addition to any special marking required by the contract, intermediate packs and shipping containers shall be marked in accordance with MIL-STD-129 or MIL-STD-1188, as applicable. Unit packs shall be identified only as to NSN, nomenclature and quantity.

6. NOTES

6.1 Intended use. The timer modification kit is used to ruggedize the time delay assembly of a cargo parachute release (M1 or M2).

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article. When a first article is required (see 3.1, 4.3 and 6.3).
(c) Selection of the applicable levels of preservation and packing (see 5.1 and 5.2).
(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2).

6.2.1 Government loaned property. The contracting officer should arrange to loan the property listed in 3.4.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample. The first article should consist of one complete timer modification kit. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

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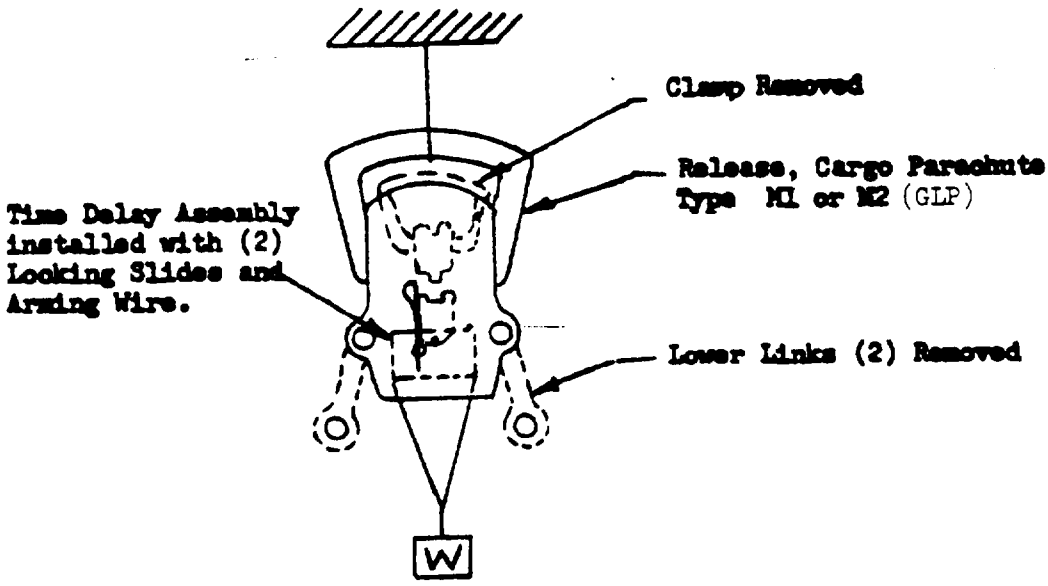
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NOTE: Total Wgt of Timer + 2 Slides + suspended Wgt -35 oz. min. (992 gm)

FIGURE - 1

ARRANGEMENT
FOR
WEIGHTED FUNCTIONAL TEST
FOR
TIME DELAY ASSEMBLY

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